# KNOWING WHEN NOT TO TRADE: PHASE 2 TECHNICAL REPORT

**Uncertainty-Aware Price Forecasting to Protect Debt-Financed Vegetable Traders in Sri Lanka**

**Domain 01: Cost of Living and Everyday Economics** · Team Model Y (Tharusha Liyanagama, Sethuli Tinara Harischandra, Janakan Jude) · IIT

## 1. Problem Statement & Economic Grounding

Small vegetable traders in Sri Lanka borrow daily working capital from informal microfinance lenders at high interest to purchase perishable produce at central wholesale hubs (Dambulla Dedicated Economic Centre) and transport it overnight to Colombo terminal markets (Peliyagoda Manning Market and Pettah). While lenders collect fixed principal and interest $(1+r)P_t^w$ regardless of market outcomes, the trader absorbs 100% of the downside selling price risk:

$$\Pi_t = P_{t+1}^r - P_t^w - C - r\,P_t^w \quad \Longleftrightarrow \quad \text{Break-Even Price: } B_t = P_t^w(1+r) + C \tag{1}$$

Conventional machine learning models output point forecasts ($R^2$ or MAE) that project false confidence. When sudden supply surges cause urban prices to crash, single-point estimates encourage reckless debt-taking, driving rural traders into insolvency. **Model Y solves this by predicting calibrated uncertainty ranges (5th, 50th, 95th percentiles), adapting under crisis distribution shifts via Adaptive Conformal Inference (ACI), detecting transmission breakdowns, and issuing a protective 3-tier signal (Green, Amber, Red) that hides point forecasts on Red days to prevent cognitive anchoring.**

## 2. Technical Architecture & Stack Rationale

- **Core Language & Modeling (Python 3.12, LightGBM, MAPIE):** LightGBM was selected over deep neural networks (e.g., TFT) because gradient-boosted trees train in seconds on tabular economic time series, handle missing values natively, and avoid overfitting on moderate sample lengths (≈3,000 daily observations).
- **Conformal Calibration Engine (CQR + ACI):** Custom 100-line implementation of Conformalized Quantile Regression and Adaptive Conformal Inference (Gibbs & Candès, 2021), mathematically validated against MAPIE (Taquet et al., 2022) with exact numerical agreement ($< 10^{-6}$ diff).
- **Storage & Data Pipeline (SQLite WAL Mode, PyMuPDF):** Local SQLite relational database (**prices.db**) ensures deterministic, reproducible walk-forward evaluation without external database network dependencies. PyMuPDF was chosen over slower OCR wrappers for its sub-second extraction of complex vector tables.
- **Delivery Interfaces (Electron Desktop + Responsive Web + Streamlit):** Dual delivery: (1) A modern, luxury agricultural dashboard built with Electron/Web matching commercial farm-management POS aesthetics; (2) An interactive multi-tab Streamlit web application for policy analysts.

## 3. Data Account, Temporality & Gap Engineering

| Source | Time Span | Extracted Volume | System Role & Handling of Missingness / Gaps |
|---|---|---|---|
| **HARTI Daily Bulletin** | 2015–2026 | 3,141 Daily PDFs | Primary target ($P_{t+1}^r$) and input features. Dashes ("–") strictly encoded as **NULL/NaN** (never zero). Multi-year layout shifts handled by modular header regex. |
| **CBSL Daily Price Report** | 2016–2026 | ≈2,500 Daily PDFs | Secondary cross-validation (Pettah/Dambulla) and supply disruption NLP flags. |
| **Open-Meteo Weather API** | 2015–2026 | 4 Agro-stations | Daily precipitation sum, 3d/7d/14d rainfall sums for Nuwara Eliya, Welimada, Matale. |
| **National Fuel / Diesel** | 2015–2026 | Revision history | Sourced from CPC/Lanka IOC to set dynamic transport cost $C_t$ based on diesel prices. |

## 4. Core Algorithmic Innovations

1. **Ratio Forecasting (Overcoming Tree Extrapolation Limits):** Rather than forecasting unbounded raw rupee prices, models predict relative price ratios $y_t = P_{t+1}^r / P_t^w$. This guarantees valid predictions even during severe inflationary regimes like 2022.
2. **Adaptive Conformal Inference (ACI) under Crisis Shocks:** Conformal prediction assumes exchangeable data, which fails during macroeconomic disruptions (e.g., 2022 fuel standstill, 80% inflation). ACI updates effective error rate $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \mathbf{1}\{y_t \notin C_t\})$ daily ($\gamma = 0.02$), dynamically expanding intervals during turbulence.
3. **Contamination-Proof Breakdown Detector:** A rolling regression $P_t^r = \beta_0 + \beta_1 P_{t-1}^w + \epsilon_t$ flags broken spatial links when residual $|\epsilon_t| > 2.5\sigma$ or $\rho_{14d} < 0.35$. Flagged days are excluded from refitting, preventing the baseline from absorbing market failures.

## 5. Honest Evaluation Against Proposal

To measure real-world performance, we conducted a rigorous **walk-forward backtest over 2,497 trading days (Oct 2019 to Oct 2026)**, strictly enforcing zero look-ahead bias: on morning $t$, only data published up to 06:00 AM on day $t$ is accessible.

### 5.1 1. Conformal Coverage: Recovery Through the 2022 Crisis

The defining test of our hypothesis is whether the model remains honest during severe economic shocks. In 2022, Sri Lanka experienced acute fuel shortages, fertilizer bans, and currency collapse:

| Regime / Year | Raw LightGBM | Static CQR | Model Y (ACI $\gamma = 0.02$) | Avg Width (% Price) | Verdict on Safety |
|---|---|---|---|---|---|
| 2020 (Lockdowns) | 71.0% | 91.0% | **91.7%** | 13.8% | Maintained target during curfews |
| 2021 (Calm) | 75.3% | 90.4% | **88.5%** | 13.5% | Narrow intervals, high precision |
| **2022 (Macro Crisis)** | **57.0% (Failed)** | **83.0% (Dipped)** | **89.3% (Recovered)** | 15.2% | **ACI restored honest 90% coverage** |
| 2023–2026 | 78.4% | 91.5% | **90.2%** | 14.1% | Stable post-crisis calibration |
| **Overall (2019–2026)** | **74.4%** | **90.5%** | **90.0%** | **14.2%** | **Exact 90.0% empirical coverage achieved** |

### 5.2 2. Financial Downside Protection & Catastrophic Loss Elimination

We simulated a trader with a standard daily purchase quota ($100$ kg) across three operational policies:

| Trading Strategy | Total Net P&L | Traded Days | Losing Days | Catastrophic ($> 30\%$) | Worst Single Day |
|---|---|---|---|---|---|
| **Always Trade** | Rs. 4,390,901 | 2,497 | 103 days | 11 events | -Rs. 6,537 |
| **Yesterday's Spread Rule** | Rs. 4,234,874 | 2,329 | 69 days | 0 events | -Rs. 3,490 |
| **Model Y (Our System)** | **Rs. 3,200,604** | **2,010** | **46 days (-55%)** | **0 events (100% avoided)** | **-Rs. 1,745 (-73%)** |

**The Economic Trade-off:** Model Y accepted Rs. 1.03M less in total profit over 7 years by refusing to trade on **487 high-risk days (RED signals)**. In exchange, it **completely eliminated catastrophic wipeouts (from 11 to 0)** and cut worst-case daily losses by 73%. For a debt-financed worker whose loan interest compounds daily, avoiding debt traps is the difference between solvency and personal ruin. In 2022 alone, Model Y issued **192 RED signals**, safely keeping traders off the road during the peak of the fuel crisis.

### 5.3 3. Planned vs. Delivered: Adherence to Contingency Cut List

- **Delivered in Full:** HARTI scraper (3,141 bulletins indexed, 2,000 price records parsed), SQLite store, LightGBM quantiles, CQR calibration, ACI adaptation, spatial breakdown detector, 3-tier decision engine, Electron/Web farm dashboard, and backtest suite.
- **Executed Contingency Cuts (per Section 8 of Proposal):**
  - *Cut 1 (TFT deep neural baseline):* Omitted to prioritize fast tabular training and explainability.
  - *Cut 2 (Multi-day horizons):* Kept strictly at $t + 1$ (dawn trading cycle).
  - *Cut 3 (SMS gateway):* Replaced by a zero-cost local Electron desktop app and responsive PWA web dashboard.
  - *Non-negotiables preserved:* Calibration, refusal logic, and walk-forward validation were 100% preserved.

## 6. Submission Artifacts & Verification Links

- **Public Git Repository:** https://github.com/Jannahan/modelx
- **Interactive Dashboard:** Run **run_desktop.bat** (Electron) or **run_web.bat** (Browser on **localhost:3000**).
- **Automated Test Suite:** Run **python tests/test_harti.py** (5/5 unit tests pass) and **python src/decision.py** (6/6 tests pass).

**Team Model Y Sign-Off:** Tharusha Liyanagama (Data & UI) · Sethuli Tinara Harischandra (Features & Models) · Janakan Jude (Calibration & Backtesting) | ModelX 2.0